Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/23/2020
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/22/2016	**End Date:**	11/21/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/21/2021 per NSSC information (Ed., 10/1/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement's material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments will satisfy two major objectives. First, the microstructure and material properties of the benchmark samples will be evaluated; different responses to thermal and mechanical loading are expected and will be characterized in detail. Any positive attributes found in the special environment will be reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or usage can have huge implications on the sustainability of the cement industry.
Task Progress:	The focus for the Microgravity Investigation of Cement Solidification (MICS) project this past year has mostly been on the samples that were a part of the Techshot Multi-use Variable-gravity Platform (MVP) experiment. The experiment had four sets of samples that hydrated at multiple gravity levels. On top of the microgravity and terrestrial gravity levels, the MVP was used to simulate the gravity levels of the moon, Mars, and a third statistical point at 70% of terrestrial gravity. The primary means of analysis so far have been through scanning electron microscopy (SEM) and image analysis. Three of the four sets of samples have been imaged. These samples include two pure compounds of ordinary portland cement and an ordinary portland cement with the lunar simulant Johnson Space Center (JSC)-1A. Task 1 – Analysis of the MVP Samples Due to the pandemic and the closure of laboratories for some time, the means of analysis have been scanning electron microscopy. This allowed for visual comparisons of how gravity influences the solidification of the samples as well as quantification of certain parameters through image analysis to gain a deeper insight. 1.1 Analysis of the Tricalcium Aluminate and Gypsum Pastes – The fractured surfaces of these samples were imaged for visual differences in the microstructure formation. The results were compared to observations from the first Microgravity Investigation of Cement Solidification (MICS) experiment. These results help further understand the dissolution of the tricalcium aluminate and gypsum and the subsequent phase formations. 1.2 Analysis of the Tricalcium Silicate Pastes – This set of samples went through an extensive SEM microstructural analysis including fractured surfaces, polished cross-sections, energy dispersive x-ray spectroscopy (EDS), and image analysis. A conference paper for the 2021 American Society of Civil Engineers (ASCE) Earth and Space conference was prepared as part of the results obtained. General trends from the microstructure were noted from the fractured and polished cross-sections. Image analysis was used to quantify the percent of each of the phases formed within the paste at all five of the gravity levels. Furthermore, the porosity was quantified through total porosity, average pore size, and the overall pore size distribution for the samples. EDS was used to see if there were any differences in the chemical composition of phases as a function of gravity level. 1.3 Analysis of the Ordinary Portland Cement with Lunar Simulant Mortars – The analysis of these samples are in the preliminary stages with fractured and polished cross-sections imaged on the SEM. More work is currently ongoing with other techniques to gain a deeper understanding into how in-situ lunar resources can play a role in a concrete material. Task 2 – Supplementary Ground Research Ground research is being done to further build off the ongoing lunar simulant study that has already resulted in one publication. The next step in the research is to maximize the use of the in-situ lunar resources and minimize the need for transporting materials to the moon from Earth. This work on developing an in-situ lunar concrete is in part supported by a NASA Space Technology Graduate Research Opportunity. The goal of the work is to create a concrete material that maximizes the use of in-situ lunar resources, be durable to withstand the harsh environment, and be able to be 3D printed. 2.1 Development of an In-Situ Lunar Concrete – Work is being done on utilizing materials similar to those found on the moon to develop an in-situ lunar concrete. The concrete proposed is different than the typical concrete that is used on Earth as it is a geopolymer concrete. This requires an aluminosilicate glassy source (lunar simulant) mixed with an alkaline solution (sodium silicate and sodium hydroxide) to initiate a polymerization process that binds the regolith together. Various solutions and lunar simulants are being explored to gain a well-rounded knowledge of the material's properties.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Abstracts for Journals and Proceedings	Collins P, Grugel R, Radlinska A. "The Influence of Variable Gravity on the Microstructural Development of Tricalcium Silicate Pastes." American Society of Civil Engineers 2020 Earth and Space Conference (postponed to 2021). American Society of Civil Engineers 2020 Earth and Space Conference, now postponed until 2021 due to Covid-19, as of April 2020. , Apr-2020
Abstracts for Journals and Proceedings	Collins P, Grugel R, Scheetz B, Radlinska A. "Microgravity's Influence on the Hydration of Tricalcium Aluminate and Gypsum Pastes." Presented at Anna Maria Workshop XX, Holmes Beach, FL, November 14, 2019. Abstracts. Anna Maria Workshop XX, Holmes Beach, FL, November 14, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Collins P. (Peter Collins) "Development of an in-situ lunar concrete." fib Young Member Group International Webinar. Online webinar. fib Young Member Group International Webinar, September 2020. , Sep-2020

Abstracts for Journals and Proceedings	Collins P. (Peter Collins) "Development of in-situ lunar concrete." Pennsylvania Aggregates and Concrete Association (PACA) online webinar, July 2020. Online webinar. Pennsylvania Aggregates and Concrete Association (PACA) online webinar, July 2020. , Jul-2020
Abstracts for Journals and Proceedings	Collins P, Edmunson J, Grugel R, Radlinska A. "Development of an In-Situ Geopolymer Concrete for Lunar Infrastructure." 2020 International Space Station Research and Development Conference (ISSRDC), Virtual Conference, September-October, 2020. 2020 International Space Station Research and Development Conference (ISSRDC), Virtual Conference, September-October, 2020. , Sep-2020
Articles in Other Journals or Periodicals	Collins PJ, Radlinska A, Grugel RN. "Cement Solidification in a Microgravity Environment." Concrete. The Magazine of the Concrete Society. 2020 Mar;54(2). , Mar-2020